

SAT Math Mastery · Trick Pack 1B

SAT Trick Pack 1 - Part B

Time	90 minutes: 70 min solving + 20 min self-check and error log
Topics	altitude to the hypotenuse · compound unit conversion · rational exponents · multi-step cost models · variance with a repeated unknown · exponential rewriting · tangency and discriminant · factoring powers · reciprocal isolation · circles and chords · margin of error · line-circle systems

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Right triangle similarity: altitude to the hypotenuse, geometric-mean relations, Pythagorean triples hidden in a trig ratio.
- S2.** Convert compound and squared units (the conversion factor is raised to the same power as the unit).
- S3.** Rewrite radicals and roots as rational exponents, merge powers of one base, raise to the reciprocal power.
- S4.** Build a function by applying several operations (fee, credit, percent change, fixed charge) in the stated order.
- S5.** Variance with an unknown repeated value: quadratic in the unknown, vertex, and the identity $\sigma^2 = \overline{x^2} - \bar{x}^2$.
- S6.** Exponential models: express a product of several bases as one base raised to $t/\text{interval}$.
- S7.** Tangency of a line and a parabola via the discriminant, then filter by a stated sign condition.
- S8.** Factor the smallest power of a variable out of a rational expression; isolate a variable that sits only in denominators.
- S9.** Circle through three points: the center lies on perpendicular bisectors; chord length from a line-circle intersection.
- S10.** Margin of error as $\propto \sqrt{p(1-p)/n}$ and ratio reasoning; line-circle system with a quadrant or sign condition.

2 · REQUIRED FACTS

- In a right triangle with altitude h to the hypotenuse, which splits it into segments p and q : $h^2 = pq$, and each $\text{leg}^2 = (\text{adjacent segment})(\text{hypotenuse})$. The three triangles are similar.
- $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$ and their multiples hide inside most trig ratios on the test.
- A unit with time^2 needs the time factor squared; a quantity measured once needs each factor once.
- $\sqrt[n]{a^m} = a^{m/n}$; $a^p a^q = a^{p+q}$; $(a^p)^q = a^{pq}$; $a^{-p} = \frac{1}{a^p}$.
- A line meets a parabola in exactly one point $\iff$ the combined quadratic has discriminant 0.
- A perpendicular bisector of a chord passes through the circle's center; all points on a circle are equidistant from the center.
- Order of operations in a billing rule matters: percent changes multiply *everything* accumulated so far.
- Margin of error $\propto \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ at a fixed confidence level; n is *inversely* proportional to the *square* of the margin.

3 · REQUIRED FORMULAS

$$h^2 = pq, \quad a^2 = pc, \quad b^2 = qc, \quad \sigma^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

$$b^2 - 4ac = 0 \text{ (tangent)}, \quad (x-h)^2 + (y-k)^2 = r^2, \quad d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M \propto \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \Rightarrow \frac{n_B}{n_A} = \frac{\hat{p}_B(1-\hat{p}_B)}{\hat{p}_A(1-\hat{p}_A)} \left(\frac{M_A}{M_B} \right)^2, \quad V = V_0 b^{t/T}$$

4 · MINI THEORY SHEET

Which trick for which item.

Item type	First move
trig ratio in a right triangle	name the triple, then set the scale from the one given length
unit with s^2 or min^2	convert length once, square the time factor
roots with a common base	write every root as a power, add exponents, undo with the reciprocal
fee / credit / percent / fixed fee	push one concrete input through every step, test the options
repeated unknown in a data set	square sum minus squared sum, then vertex (mean of the others)
several bases, one interval	prime-factor, add exponents, divide by the interval length
tangent or exactly one	set the combined discriminant to 0, then apply the sign condition
x only in denominators	cancel common factors, solve for $\frac{1}{x}$, flip last
chord of a circle	center on the bisector; equal distances give one linear equation

Speed shortcuts.

Situation	Shortcut
altitude h to hypotenuse	$h^2 = pq$ and $\text{leg}^2 = (\text{segment})(\text{hypotenuse})$; scale a 3-4-5 or 5-12-13
$\text{ft/s}^2 \rightarrow \text{mi/min}^2$	multiply by 60^2 (time) and divide by 5280 (distance)
$\sqrt[3]{x} \cdot \sqrt[4]{x^3} \dots$	add the rational exponents, then raise to the reciprocal
multi-step bill	evaluate $f(0)$ and $f(1)$ in order; slope and intercept follow
repeated unknown x in data	minimum variance at $x = \text{mean of the other values}$
$b_1^{t/a} b_2^{t/c}$	exponent of 2 per interval, e.g. $2^{5t/6} = (2^5)^{t/6}$
tangent line to parabola	combine, $b^2 - 4ac = 0$, solve, <i>then</i> check the stated sign
$\frac{k(c-3)}{x(c-3)}$	cancel $(c-3)$ immediately
chord with equal y -values	center's x is the midpoint of the chord's x -values
margin of error halves	n multiplies by 4; extra factor from $\hat{p}(1-\hat{p})$ ratio
line meets circle	substitute the line; the roots give both points; apply the quadrant condition last

5 · GUIDED EXAMPLES

Example 1 — Squared time unit

A cyclist accelerates at 0.5 m/s^2 . Express this in km/min^2 .Length: $1 \text{ m} = \frac{1}{1000} \text{ km}$ (once). Time: $1 \text{ s} = \frac{1}{60} \text{ min}$, so $\frac{1}{\text{s}^2} = 60^2 \cdot \frac{1}{\text{min}^2}$ (squared).

$$0.5 \cdot \frac{3600}{1000} = 1.8 \text{ km/min}^2.$$

Example 2 — Roots to rational exponents

For $x > 0$, $\sqrt{x} \cdot \sqrt[3]{x^2} = 5^{7/3}$. Find x . $x^{1/2} \cdot x^{2/3} = x^{7/6}$. So $x^{7/6} = 5^{7/3}$; raise both sides to $\frac{6}{7}$: $x = 5^{\frac{7}{3} \cdot \frac{6}{7}} = 5^2 = 25$.

Example 3 — Tangent line and a sign condition

The line $y = mx + 4$ touches the parabola $y = x^2 + 5$ at exactly one point, and $m > 0$. Find m . $x^2 + 5 = mx + 4 \Rightarrow x^2 - mx + 1 = 0$. Discriminant $m^2 - 4 = 0 \Rightarrow m = \pm 2$. The condition $m > 0$ leaves $m = 2$.

6 · PRACTICE SET — 15 PROBLEMS · ALL SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. SAT CHALLENGE · NO CALCULATOR NEEDED
In triangle PQR , the angle at Q is a right angle. Point S lies on side $\overline{PR}$ so that $\overline{QS} \perp \overline{PR}$. If $\tan \angle QRP = \frac{4}{3}$ and $PS = 16$, what is the length of $\overline{SR}$?

(A) 25

(B) 9 ✓

(C) 12

(D) 15

- $\tan \angle QRP = \frac{PQ}{QR} = \frac{4}{3}$, so the legs are in the ratio 4 : 3 and the triangle is a 3-4-5 triangle: $PQ = 4k$, $QR = 3k$, $PR = 5k$.
- Altitude relation $PQ^2 = PS \cdot PR$: $16k^2 = 16 \cdot 5k \Rightarrow k = 5$. So $PR = 25$.
- $SR = PR - PS = 25 - 16 = 9$.
- Check: $QR^2 = SR \cdot PR$ gives $15^2 = 9 \cdot 25$. The other values are $QS = 12$, $QR = 15$, $PR = 25$, which are different segments.

Answer: (B)

2. SAT CHALLENGE · CALCULATOR
A test vehicle accelerates at a constant 22 feet per second squared. Which of the following is equal to this acceleration in miles per minute squared? (1 mile = 5280 feet)

(A) $\frac{1}{4}$ (B) $\frac{1}{240}$

(C) 15 ✓

(D) 79,200

- Distance: $1 \text{ ft} = \frac{1}{5280} \text{ mi}$, used once.
- Time: $1 \text{ s} = \frac{1}{60} \text{ min}$, so $\frac{1}{\text{s}^2} = 60^2 \cdot \frac{1}{\text{min}^2}$; the time factor is squared.
- $22 \cdot \frac{3600}{5280} = \frac{79,200}{5280} = 15$ miles per minute squared.
- Check by speeds: after 1 min the speed is $22 \cdot 60 = 1320 \text{ ft/s} = 15 \text{ mi/min}$.

Answer: (C)

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive real number x , $\frac{\sqrt[3]{x} \cdot \sqrt[4]{x^3}}{\sqrt[6]{x}} = \sqrt[3]{k}$, where $k = 2^{11}$. Which of the following is equal to x ?

(A) 2^4 ✓

(B) $2^{\frac{11}{3}}$

(C) $2^{\frac{121}{36}}$

(D) $2^{\frac{44}{3}}$

1. Rational exponents: $x^{1/3} \cdot x^{3/4} \cdot x^{-1/6}$.

2. Add: $\frac{1}{3} + \frac{3}{4} - \frac{1}{6} = \frac{4+9-2}{12} = \frac{11}{12}$, so the left side is $x^{11/12}$.

3. Right side: $\sqrt[3]{2^{11}} = 2^{11/3}$.

4. $x^{11/12} = 2^{11/3}$. Raise both sides to the reciprocal power $\frac{12}{11}$: $x = 2^{\frac{11}{3} \cdot \frac{12}{11}} = 2^4$.

Answer: (A)

4.

SAT CHALLENGE · CALCULATOR

A utility computes a customer's monthly water bill B , in dollars, for u units of water in this order: a service fee of \$18 plus \$2.40 for each unit used; then a one-time credit of \$12 is subtracted; then a 5% surcharge is added to the amount at that point; and finally a fixed \$3.50 fee is added. Which of the following defines B for u units?

(A) $B = 2.52u + 10.4$

(B) $B = 2.52u + 9.975$

(C) $B = 2.52u + 22.4$

(D) $B = 2.52u + 9.8$ ✓

1. After fee and usage: $18 + 2.4u$. After the credit: $6 + 2.4u$.

2. A 5% surcharge multiplies this amount by 1.05: $1.05(6 + 2.4u) = 6.3 + 2.52u$.

3. The fixed fee is added last, after the surcharge: $B = 2.52u + 6.3 + 3.5 = 2.52u + 9.8$.

4. Check with $u = 10$: $18 + 24 = 42$, $42 - 12 = 30$, $30 \cdot 1.05 = 31.5$, $31.5 + 3.5 = 35$, and $2.52(10) + 9.8 = 35$.

Answer: (D)

5.

SAT CHALLENGE · CALCULATOR

A data set consists of the five values 5, 8, 17, x , x , where x is a real number. The variance of a data set is the mean of the squared deviations of its values from their mean. What is the least possible value of the variance of this data set?

(A) 26

(B) 78

(C) 15.6 ✓

(D) 10

$$1. \sum v = 30 + 2x \text{ and } \sum v^2 = 378 + 2x^2, \text{ so } \sigma^2 = \frac{378 + 2x^2}{5} - \left(\frac{30 + 2x}{5}\right)^2.$$

$$2. \text{ Expand: } \sigma^2 = \frac{1890 + 10x^2 - 900 - 120x - 4x^2}{25} = \frac{6x^2 - 120x + 990}{25}.$$

$$3. \text{ This quadratic opens upward; its vertex is at } x = \frac{120}{12} = 10, \text{ the mean of } 5, 8, 17.$$

$$4. \sigma_{\min}^2 = \frac{600 - 1200 + 990}{25} = \frac{390}{25} = 15.6. \text{ (Check: mean} = 10, \frac{25 + 4 + 49 + 0 + 0}{5} = 15.6.)$$

Answer: (C)

6.

SAT CHALLENGE · CALCULATOR

The size of a bacterial colony, in thousands of cells, is modeled by $V(t) = 500 \cdot 2^{t/2} \cdot 4^{t/3} \cdot 16^{-t/12}$, where t is the number of hours since the start of observation. The size of the colony is multiplied by a constant factor c every 6 hours. What is the value of c ?

$$1. \text{ Write every base as a power of 2: } 4^{t/3} = 2^{2t/3} \text{ and } 16^{-t/12} = 2^{-t/3}.$$

$$2. \text{ Add the exponents: } \frac{t}{2} + \frac{2t}{3} - \frac{t}{3} = \frac{3t + 4t - 2t}{6} = \frac{5t}{6}, \text{ so } V(t) = 500 \cdot 2^{5t/6}.$$

$$3. \text{ Rewrite with the interval in the exponent: } 2^{5t/6} = (2^5)^{t/6} = 32^{t/6}.$$

$$4. c = 32. \text{ (Check: } V(t+6)/V(t) = 2^5 = 32.)$$

Answer: 32

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = mx - 5$ intersects the parabola $y = x^2 - 6x + 11$ at exactly one point, where m is a negative constant. What is the value of m ?

(A) 14

(B) -14 ✓

(C) -2

(D) 2

1. Set the expressions equal: $x^2 - 6x + 11 = mx - 5 \Rightarrow x^2 - (6 + m)x + 16 = 0$.
2. Exactly one intersection means discriminant 0: $(6 + m)^2 - 4(1)(16) = 0 \Rightarrow (6 + m)^2 = 64$.
3. $6 + m = 8$ or $6 + m = -8$, so $m = 2$ or $m = -14$.
4. The constant m must be negative, so $m = -14$. (Then $x = \frac{6 + m}{2} = -4$, and both sides equal 51 there.)

Answer: (B)

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

For $x > 0$ and any positive integer n , which of the following is equivalent to $\frac{(x^{2n+1})^3 + 4x^{6n+3} - 2x(x^{3n+1})^2}{x^{6n+1}}$?

(A) $5x^2$ (B) $3x^4$ (C) $3x^{12n+4}$ (D) $3x^2$ ✓

1. Power of a power: $(x^{2n+1})^3 = x^{6n+3}$ and $x(x^{3n+1})^2 = x \cdot x^{6n+2} = x^{6n+3}$.
2. The numerator is $x^{6n+3}(1 + 4 - 2) = 3x^{6n+3}$.
3. Divide, subtracting exponents: $3x^{(6n+3)-(6n+1)} = 3x^2$.

Answer: (D)

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers $k \neq 3$ and $x \neq 0$:

$$\frac{4(k-3)}{x(k-3)} + \frac{2}{x} - \frac{3}{4} = \frac{3}{8}$$

What is the value of x ?

1. Cancel the common factor $(k-3)$: $\frac{4}{x} + \frac{2}{x} - \frac{3}{4} = \frac{3}{8}$.

2. Combine: $\frac{6}{x} = \frac{3}{8} + \frac{3}{4} = \frac{9}{8}$.

3. Solve for $\frac{1}{x}$: $\frac{1}{x} = \frac{9}{8 \cdot 6} = \frac{3}{16}$.

4. Flip at the end: $x = \frac{16}{3}$.

Answer: $\frac{16}{3}$ **10.**

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, a circle passes through the points $(-1, 2)$, $(7, 2)$ and $(6, 3)$. Which of the following is an equation of the circle?

(A) $(x-3)^2 + (y+1)^2 = 25$ ✓

(B) $(x-3)^2 + (y-2)^2 = 16$

(C) $(x-3)^2 + (y+1)^2 = 5$

(D) $(x-3)^2 + (y-1)^2 = 17$

1. The points $(-1, 2)$ and $(7, 2)$ form a horizontal chord, so the center lies on the vertical line $x = \frac{-1+7}{2} = 3$. (It is *not* the midpoint $(3, 2)$; the chord need not be a diameter.)

2. Let the center be $(3, k)$. Equal distances to $(7, 2)$ and $(6, 3)$: $16 + (2-k)^2 = 9 + (3-k)^2 \Rightarrow 20 - 4k = 18 - 6k \Rightarrow k = -1$.

3. $r^2 = (7-3)^2 + (2+1)^2 = 16 + 9 = 25$.

4. Equation: $(x-3)^2 + (y+1)^2 = 25$. Check $(6, 3)$: $9 + 16 = 25$.

Answer: (A)

11.

SAT CHALLENGE · NO CALCULATOR NEEDED

A rectangular banner has an area of 540 square inches. The length of the banner is 150% of its width, and the width is $c\sqrt{10}$ inches, where c is a positive constant. What is the value of c ?

1. Length = $1.5 \cdot \text{width} = \frac{3}{2} c\sqrt{10}$.
2. Area = length $\cdot$ width = $\frac{3}{2} (c\sqrt{10})^2 = \frac{3}{2} \cdot 10c^2 = 15c^2$.
3. $15c^2 = 540 \Rightarrow c^2 = 36$, and since $c > 0$, $c = 6$.

Answer: 6**12.**

SAT CHALLENGE · CALCULATOR

A crew spreads fertilizer on two kinds of turf. Type A turf receives 30 grams per square meter, and Type B turf receives 4 kilograms per 100 square meters. In all, the crew treats 70,000 square meters and uses 2.6 metric tons of fertilizer (1 metric ton = 1000 kilograms). How many square meters of Type A turf are treated?

1. Put both rates in kilograms per square meter: Type A = $\frac{30}{1000} = 0.03$; Type B = $\frac{4}{100} = 0.04$. Total fertilizer = $2.6 \times 1000 = 2600$ kg.
2. Let a be the area of Type A, so Type B has $70,000 - a$: $0.03a + 0.04(70,000 - a) = 2600$.
3. $2800 - 0.01a = 2600 \Rightarrow a = 20,000$.
4. Check: $0.03(20,000) + 0.04(50,000) = 600 + 2000 = 2600$ kg.

Answer: 20000

13.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $x + y = 4$ intersects the circle $x^2 + y^2 - 4x + 2y = 24$ at two points. What is the distance between these two points?

(A) 7

(B) $7\sqrt{2}$ ✓

(C) 14

(D) $\frac{7\sqrt{2}}{2}$

1. Substitute $y = 4 - x$: $x^2 + (4 - x)^2 - 4x + 2(4 - x) = 24$.
2. Expand: $2x^2 - 8x + 16 - 4x + 8 - 2x = 24 \Rightarrow 2x^2 - 14x = 0 \Rightarrow 2x(x - 7) = 0$.
3. $x = 0$ gives $(0, 4)$; $x = 7$ gives $(7, -3)$.
4. Distance = $\sqrt{(7 - 0)^2 + (-3 - 4)^2} = \sqrt{98} = 7\sqrt{2}$. (The horizontal difference 7 alone is not the distance, and no halving is needed.)

Answer: (B)

14.

SAT CHALLENGE · CALCULATOR

Two polls use the same confidence level. For a poll of n people with sample proportion $\hat{p}$, the margin of error is proportional to $\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$. Poll A surveyed 625 people and had $\hat{p} = 0.50$. Poll B had $\hat{p} = 0.10$ and a margin of error half as large as the margin of error of Poll A. How many people did Poll B survey?

1. Squaring $M \propto \sqrt{\hat{p}(1 - \hat{p})/n}$ gives $n \propto \frac{\hat{p}(1 - \hat{p})}{M^2}$.
2. Ratio: $\frac{n_B}{n_A} = \frac{0.10 \cdot 0.90}{0.50 \cdot 0.50} \cdot \left(\frac{M_A}{M_B}\right)^2 = \frac{0.09}{0.25} \cdot 2^2 = 0.36 \cdot 4 = 1.44$.
3. $n_B = 625 \cdot 1.44 = 900$.

Answer: 900

15.

SAT CHALLENGE · DESMOS

$$x^2 + y^2 = 65$$

$$x - y = 7$$

The solution (a, b) to the system of equations above satisfies $a > 0$. What is the value of $a + 3b$?

1. Write $y = x - 7$ and substitute: $x^2 + (x - 7)^2 = 65 \Rightarrow 2x^2 - 14x - 16 = 0 \Rightarrow x^2 - 7x - 8 = 0$.
2. Factor: $(x - 8)(x + 1) = 0$, so $x = 8$ or $x = -1$.
3. $x = 8$ gives $y = 1$; $x = -1$ gives $y = -8$. Only $(8, 1)$ has $a > 0$.
4. $a + 3b = 8 + 3(1) = 11$.

Answer: 11